



Alryada university
For science and technology



Lecture 3

Bacterial Physiology

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2025 – 2026

Lecture Learning Outcomes

By the end of this lecture, the student will be able to:

1. **Define** bacterial physiology (growth, nutrition, metabolism).
2. **Identify** basic requirements for bacterial growth.
3. **Classify** bacterial products (enzymes, pigments, toxins).
4. **Relate** bacterial toxins to associated diseases.

What is Bacterial Physiology?

Definition: Study of how bacteria **live, grow, and survive.**

Main Aspects:

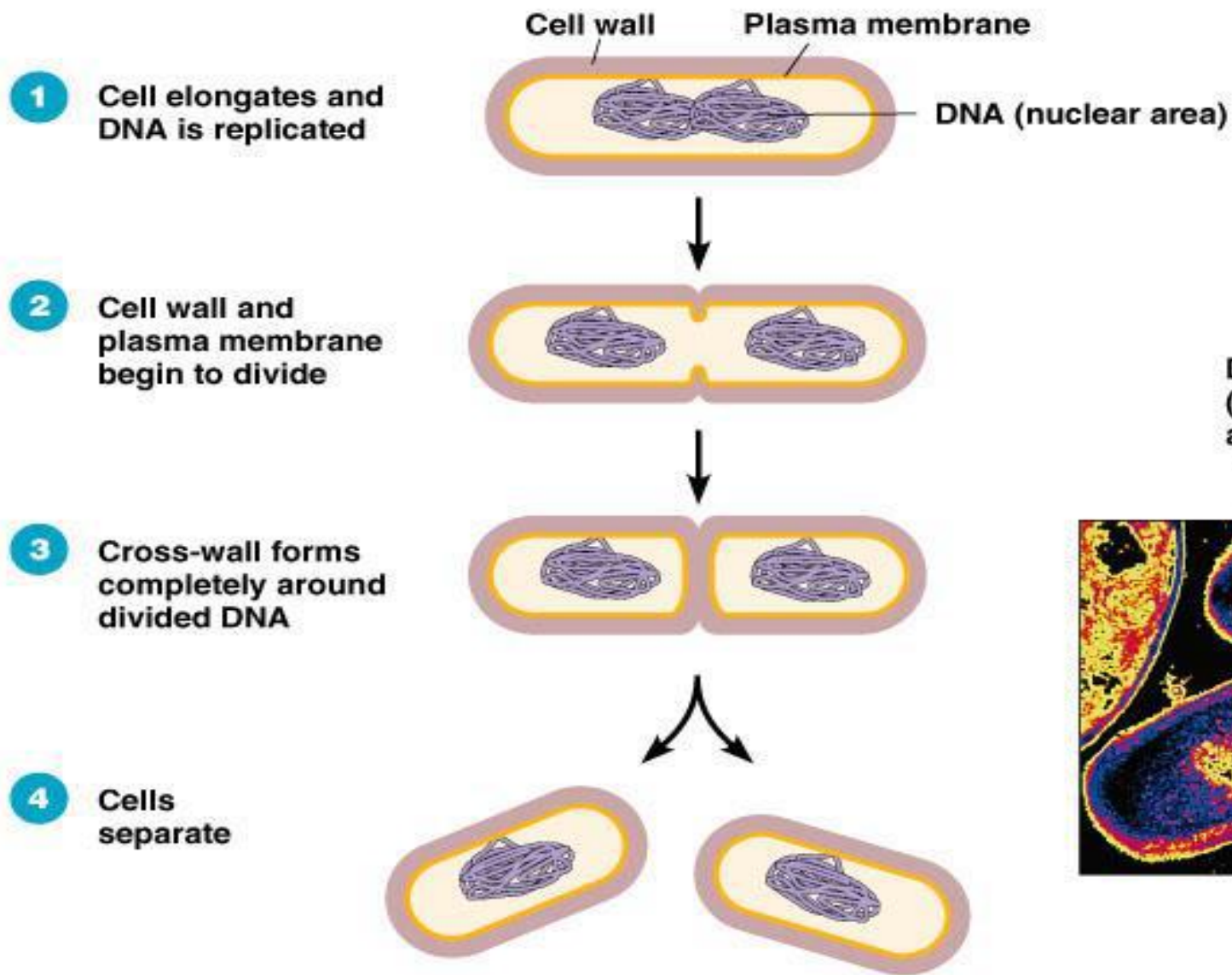
- **Bacterial growth**
- **Nutrition**
- **Metabolism**

Bacterial Growth????

Refers to the number of cells or the size???

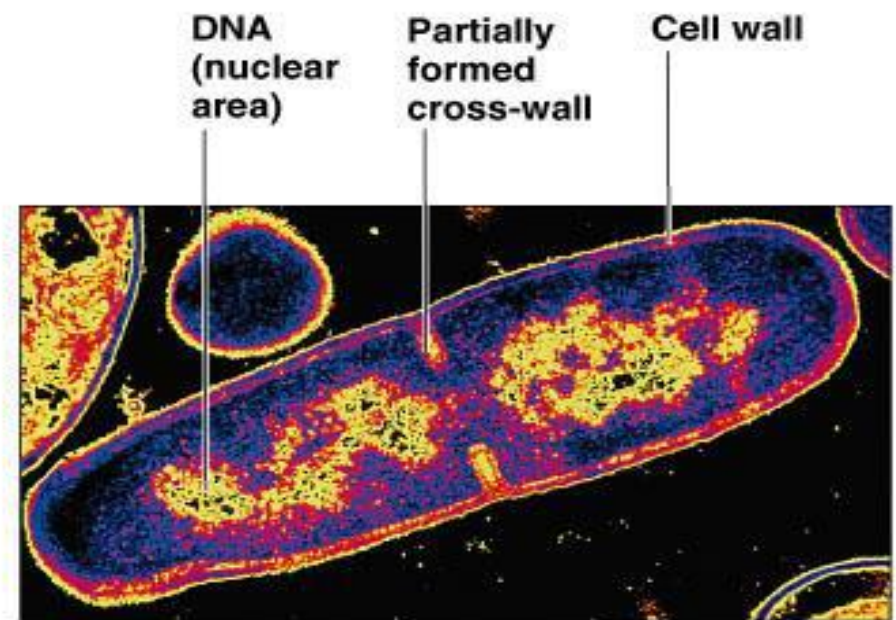
- Increase in the numbers of cells NOT cell size

➤ One cell becomes **colony** of **millions of cells**



(a) A diagram of the sequence of cell division.

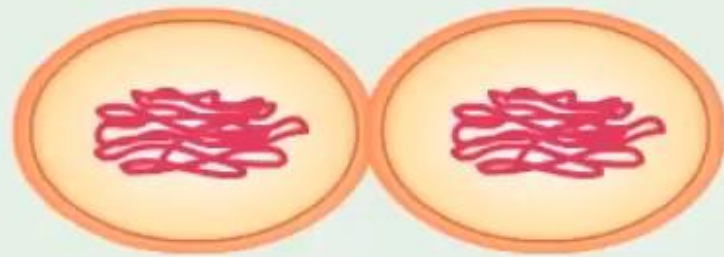
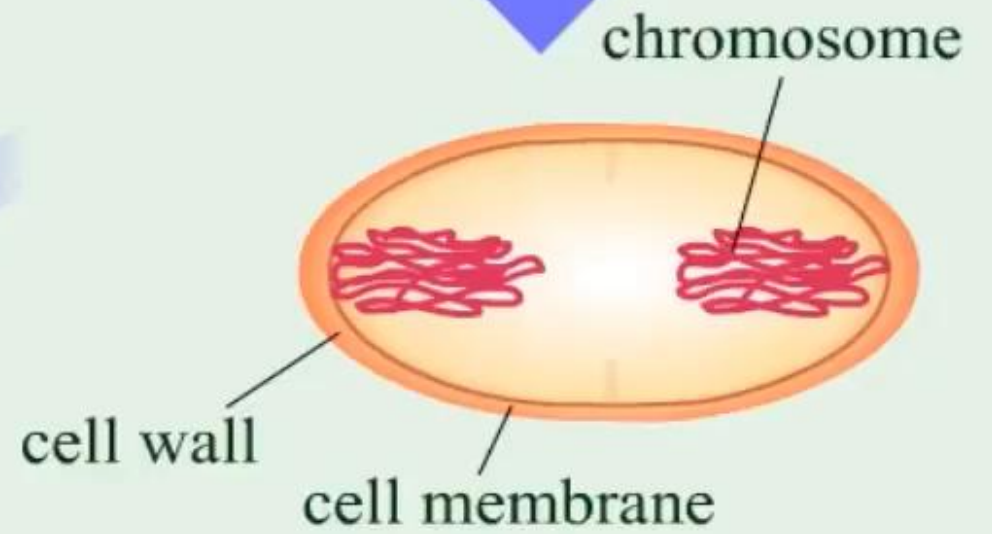
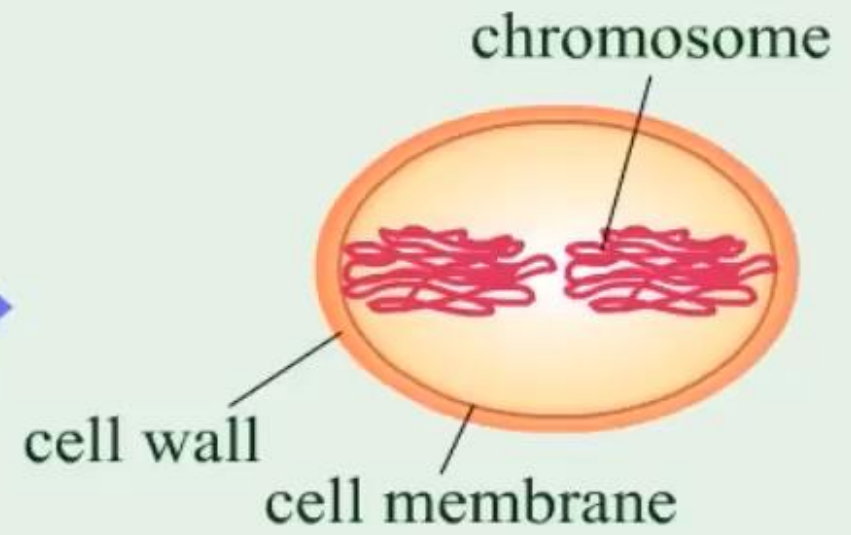
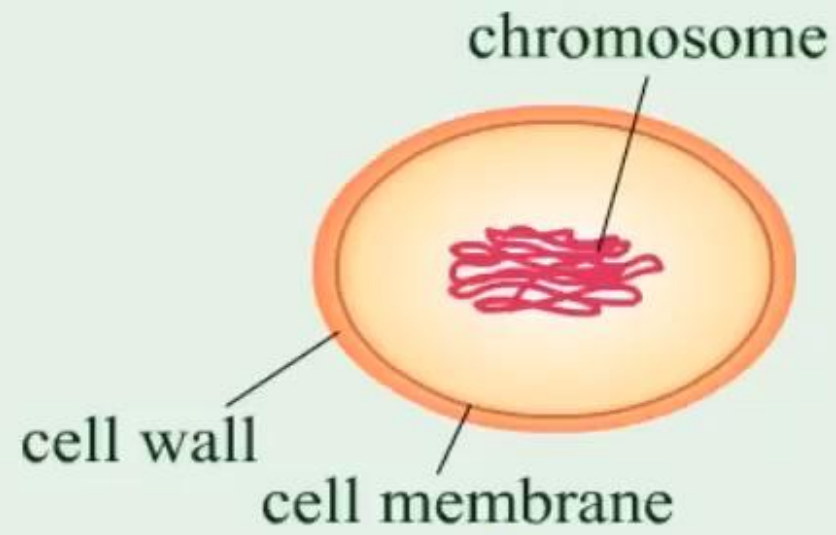
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(b) A thin section of a cell of *Bacillus licheniformis* starting to divide.

Binary Fission – How One Bacterium Becomes Two??

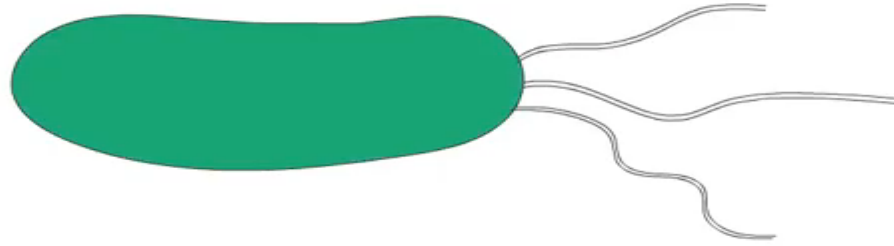
Watch this short video to understand the mechanism of bacterial cell division.



Generation Time

- Time required for cell to divide/for population to double
- Average for bacteria is 1-3 hours
- *E. coli* generation time = 20 min
 - 20 generations (7 hours), 1 cell becomes 1 million cells!

In this short video, you will learn what “generation time” means — the time it takes for one bacterial cell to divide and produce two identical daughter cells.



Interactive Questions

Question 1 – Concept Understanding

Q: What does generation time mean in bacteria?

- a) The total life span of a bacterial cell
- b) The time it takes for one cell to divide into two
- c) The duration of the stationary phase
- d) The time needed for spore formation

Question 2 – Application

Q: If the generation time of E. coli is 20 minutes, how many cells will result from one cell after 1 hour (under ideal conditions)?

- a) 2 cells
- b) 4 cells
- c) 8 cells
- d) 16 cells

Question 3 – Critical Thinking

Q: Why do pathogenic bacteria with shorter generation times cause more severe infections?

- a) They produce fewer toxins
- b) They multiply faster and overwhelm host defenses
- c) They die quickly inside the host
- d) They require more nutrients

How can we detect that bacterial growth actually occurred inside the body?

A. Through Clinical Signs

(Fever ,Inflammation , redness, pain, Fatigue, loss of appetite, rapid heartbeat)

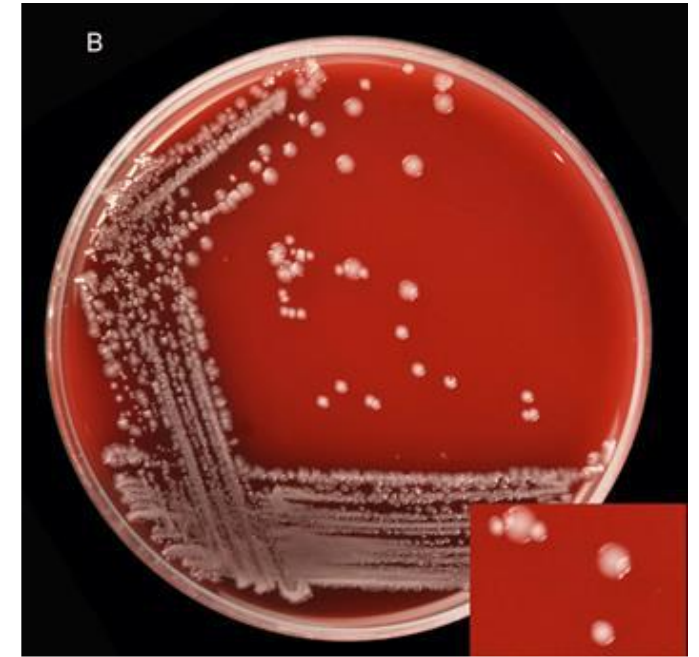
B. Through Laboratory Tests

A sample (blood, urine, sputum, wound swab, etc.)

Measuring growth

1. Culture Test:

2. Microscopic Examination



OUR AGENDA

- **Growth requirements.**
- **Bacterial products.**



Bacterial Requirements for Growth

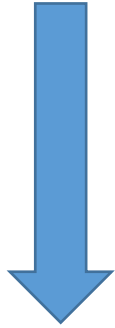
Bacteria need **food** + proper **environment** to grow.

Nutrients → Carbon, Nitrogen, Water, Minerals

Conditions → Temperature, pH, Oxygen, Moisture



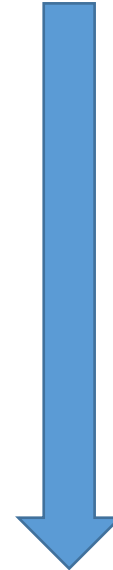
Nutrients



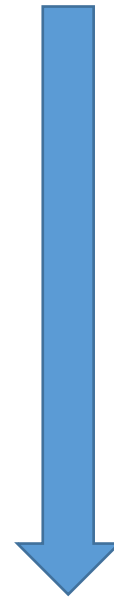
Gases



Moisture

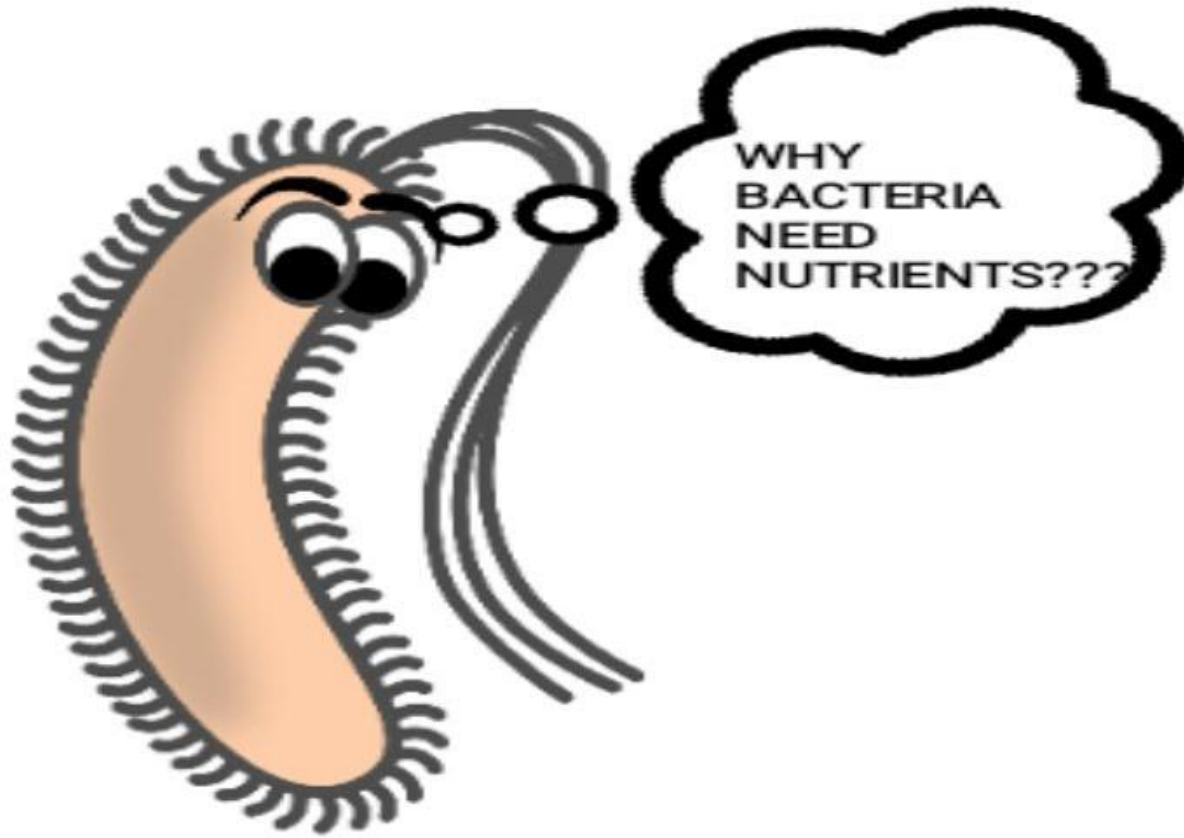


Temperature



PH

Nutrition Of Bacteria



- 1- To make carbohydrate
- 2-To make amino acids
- 3- To make lipids , vitamins, growth factors
- 4-To make RNA/ DNA

Autotrophs:

Can synthesize all their organic compounds by utilizing atmospheric CO₂, N₂ (**NO medical importance**)

•Heterotrophs

- Unable to synthesize their own metabolites
- Depend on preformed organic compounds
- Includes **all pathogenic bacteria**

Mode of nutrition in Bacteria



Nutritional requirements

1- Carbon (carbohydrates, lipids) → gives energy and structure

2- N (amino acids, proteins) → builds proteins and genetic material

3- GROWTH factors

ready-made organic compounds

→ Needed by some bacteria because they can't synthesize them.

- ✓ Include: vitamins, amino acids, nucleotides
- ✓ Help in enzyme function, DNA synthesis, and cell growth..

Gaseous requirements

02

obligate
aerobes



A

obligate
anaerobes



B

facultative
anaerobes



C

microaerophiles:



D

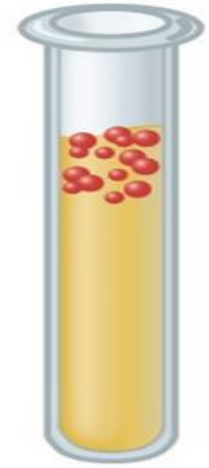
Gaseous requirements

Obligatory aerobes:

➤ Grow only in the presence of O₂.

e.g. T.B and *Pseudomonas*.

obligate
aerobes



facultative anerobes:

➤ Grow well in the presence of O₂, as well as its absence.

E.g. most pathogenic bacteria as *Staphylococci*, and *E. coli*.

facultative
anaerobes



Gaseous requirements

Obligatory anerobes:

- Grow only in **absence of O₂** as they lack peroxidase enzyme.

E.g. Clostridium.

obligate
anaerobes



microaerophiles



Microaerophilic :

- Grow only in the presence of **trace amount of O₂**

E.g: H.pylori

CO₂

- Capnophilic

bacteria that require higher levels of carbon dioxide (CO₂) usually between 5 - 10% for their optimal growth.

- Ex. *Neisseria meningitides*
- *Neisseria gonorrhoeae*



MOISTURE

the amount of **water** available in the **environment** or inside the bacterial cell.

- Essential for biochemical reactions inside the bacterial cell (metabolism and transport).
- Helps in the transport of nutrients and waste across the cell membrane
- Maintains cell shape and stability, preventing dehydration.
- **Lack of moisture** leads to growth inhibition or **bacterial death**.
- **Some bacteria form spores to survive in dry conditions.**



TEMPERATURE

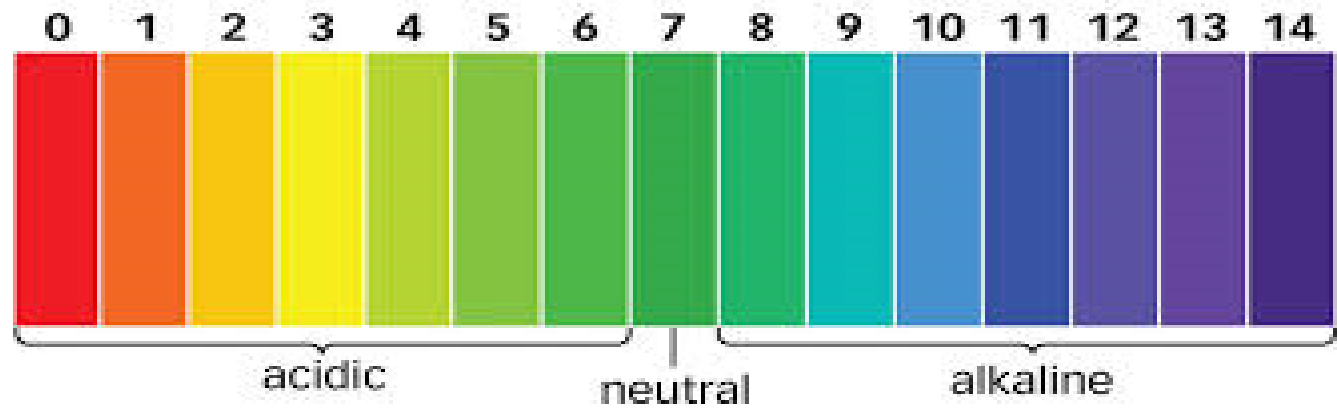
- 37 °C
- Mesophile
(20-45 °C)



- Most pathogenic bacteria are **mesophiles**
- Grow best at 37°C (human body temperature)
- Growth range 20–45°C

PH

- 7.5 (7.2-7.6)



Lactobacilli

Vibrio cholera

OUR AGENDA

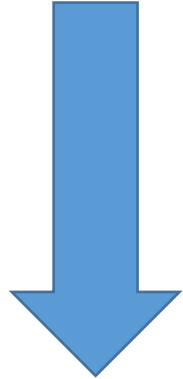
- **Growth requirements.**
- **Bacterial products.**



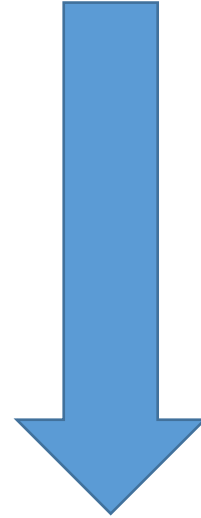
Bacterial products



Enzymes



Pigments



Toxins

1-ENZYMES

- Enzymes are **biological catalysts** that increase the rate of a chemical reaction
- The enzyme is not altered or consumed in the reaction
- The substance which is acted upon by an enzyme is called **a substrate.**

FUNCTIONS

1.Nutritional

2.Virulence

3.Antibiotic resistance

4.Laboratory diagnosis



Nutritional

Some bacteria produce enzymes that help them **digest complex nutrients**, enabling **survival** in different environments.

Virulence

Genes and enzymes that increase a microbe's ability to **cause disease**, such as **toxins** or **tissue-destroying enzymes**.

Antibiotic resistance

Certain bacteria carry **resistance genes**, allowing them to survive antibiotic treatment.

Laboratory diagnosis

Enzymes and resistance patterns are used in labs **to identify** microorganisms and guide **proper treatment**.

2-Bacterial pigments

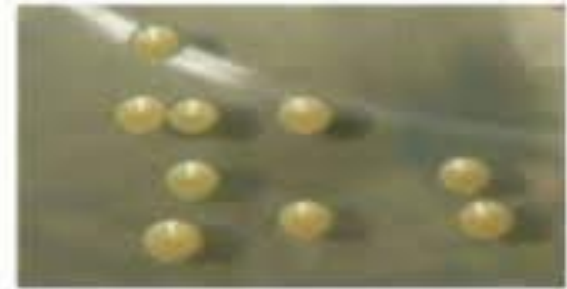


Bacterial pigments

Bacterial pigments are colored substances produced by certain bacteria as secondary metabolites.

Types

- **Pigment production:**
 - **Endopigment** (restricted to the colonies):
 - Golden yellow with *Staphylococcus aureus*.
 - White with *Staph. epidermidis*.
 - **Exopigment** (the color diffuses in the surrounding medium):
 - Green exopigment with *Pseudomonas aeruginosa*.



FUNCTIONS

IDENTIFICATION



3-BACTERIAL TOXINS

Toxin Production

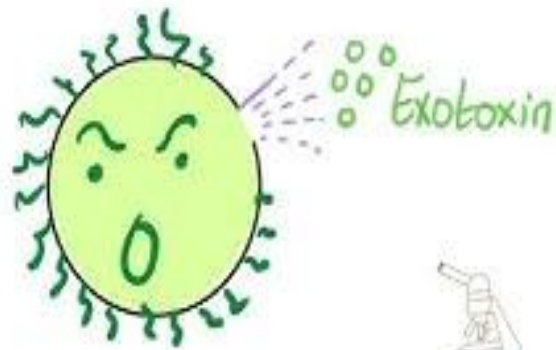
Toxins: Poisonous substances produced by microbes and lead to body damage.

Two **types** of toxins: **Exotoxins**

Endotoxins

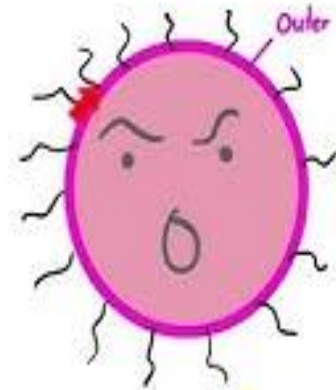


Exotoxin



VS

Endotoxin



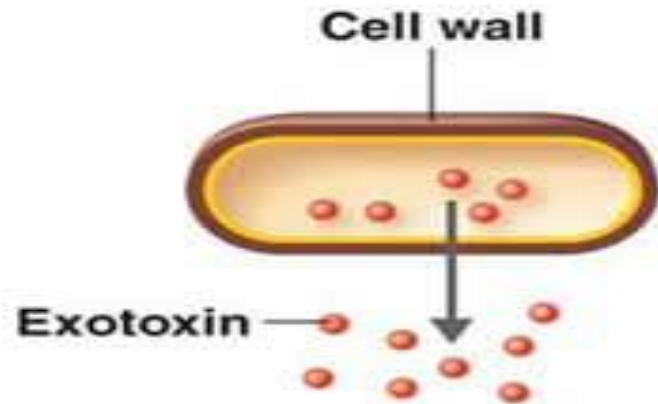
Microbiology &
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Medicus

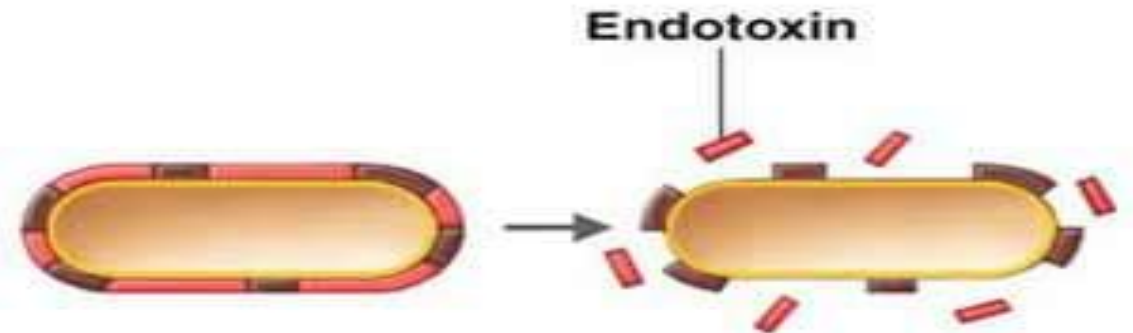


Exotoxins Vs Endotoxins

Differences Between Exotoxins and Endotoxins



(a) Exotoxins are proteins produced inside pathogenic bacteria, most commonly gram-positive bacteria, as part of their growth and metabolism. The exotoxins are then secreted or released into the surrounding medium following lysis.



(b) Endotoxins are the lipid portions of lipopolysaccharides (LPSs) that are part of the outer membrane of the cell wall of gram-negative bacteria (lipid A; see Figure 4.13c). The endotoxins are liberated when the bacteria die and the cell wall breaks apart.

Comparison Between Exotoxins and Endotoxins

Feature	Exotoxins	Endotoxins
Chemical nature	Proteins	Lipid (Lipid A of LPS)
Produced by	Mostly Gram-positive bacteria	Gram-negative bacteria only
Location	Secreted outside the cell	Part of outer membrane
Release	Actively secreted during growth	Released when bacteria die (cell lysis)
Heat stability	Heat-labile (destroyed by heat)	Heat-stable
Toxicity	Very high toxicity	Moderate toxicity
Specificity	Specific action (target specific cells)	Non-specific (general inflammatory response)
Antitoxin formation	Yes (can form toxoid vaccines)	No toxoid formation

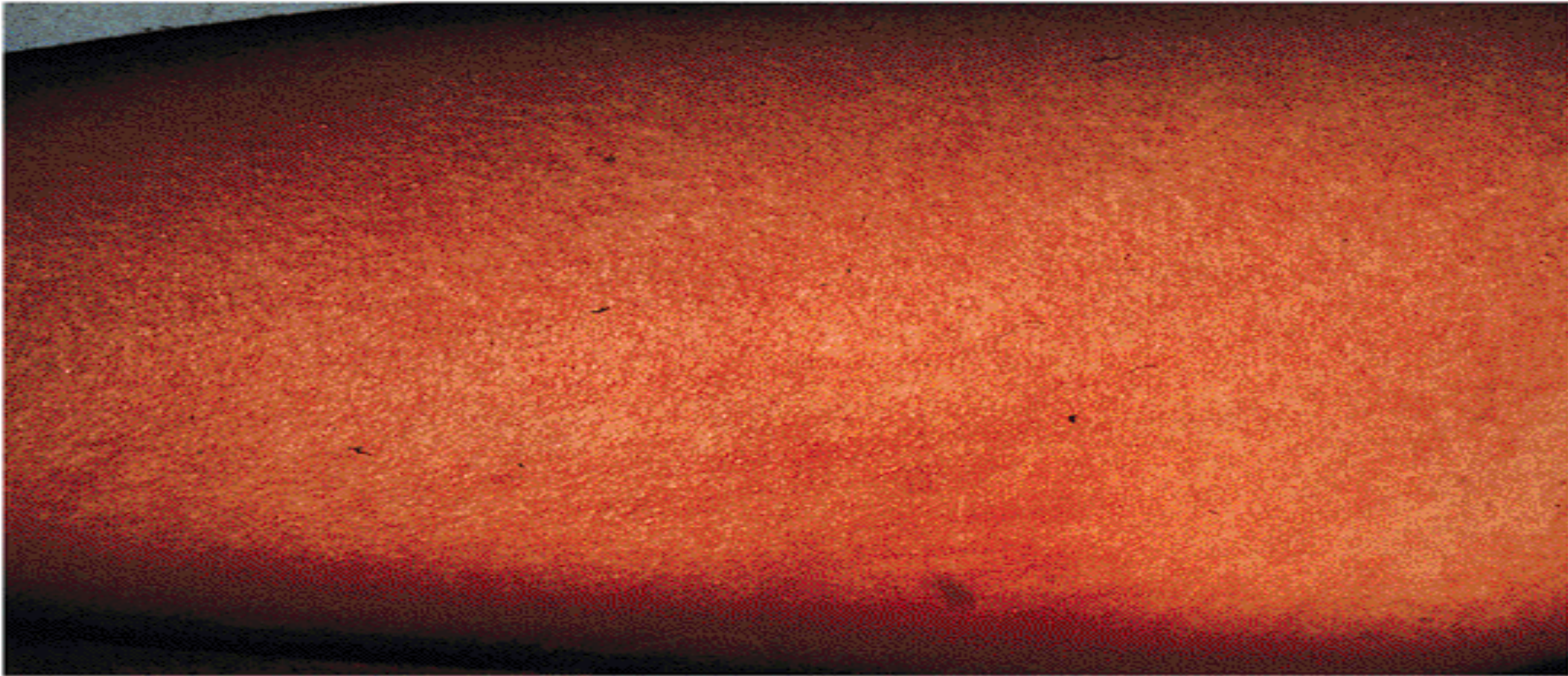
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1-Muscle Spasms of Tetanus are Caused by Neurotoxin of Clostridium tetani



Neonatal Tetanus (Wrinkled brow and risus sardonicus)

2-Rash of Scarlet Fever Caused by Erythrogenic Toxins of *Streptococcus pyogenes*



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3-Diseases Caused by Staphylococcal Toxins



Scalded Skin Syndrome



Toxic Shock Syndrome

Uses of Exotoxins

1)Preparation of antitoxins for passive immunization

- Preparation of **Antitoxins** for Passive Immunization
- Antitoxins are **ready-made antibodies** that neutralize bacterial toxins.
- They provide immediate but short-term protection.

2)Preparation of toxoid for active immunization

- Preparation of **Toxoid for Active Immunization**
- **Toxoid** is an inactivated toxin used as a **vaccine**.
- It stimulates the immune system to produce long-lasting immunity.



The time required for bacterial population to double is called:

- a) Lag time b) Generation time c) Metabolism
 - c) Stationary phase d) Metabolism
-

Exotoxins are mainly:

- a) Lipids b) Proteins
 - c) Carbohydrates d) DNA
-

Endotoxin is part of:

- a) Gram positive cell wall b) Gram negative outer membrane
 - c) Capsule d) Cytoplasm
-

Bacterial pigments are considered:

- a) Primary metabolites b) Secondary metabolites
 - c) Enzymes d) Growth factors
-

All pathogenic bacteria are:

- a) Autotrophs b) Heterotrophs c) Capnophilic
- c) Heterotrophs d) Obligate anaerobes

dr. rania hamed

➤ **Compare between ENDO AND EXOTOXINS ?**

➤ **A patient develops muscle spasms after a contaminated wound.**

- **What type of toxin is responsible?**
- **Is it exotoxin or endotoxin?**

Thank You

